

Chua Han Chong

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Education

- PhD** **Nanyang Technological University**, Computer Science Singapore
• Research areas: spatial intelligence, embodied AI, real2sim2real.
• Supported by the NTU Research Scholarship. Aug 2026 – present
- MSc** **Nanyang Technological University**, Artificial Intelligence Singapore
• GPA: 4.30/5.00. Aug 2023 – Jan 2026
- BEng** **Universiti Putra Malaysia**, Mechanical Engineering Kuala Lumpur, Malaysia
• CGPA: 3.728/4.00. Sept 2013 – July 2017

Experience

- Porolis Technologies Pte. Ltd.**, AI Engineer Lead Singapore
May 2023 – Aug 2026
3 years 4 months
- Architected and deployed the production AI platform serving all gemstone analysis models (FastAPI, Redis, RQ, Docker, nginx, AWS EC2 and S3), covering API gateway and load balancing, distributed job queues with priority-based worker scheduling, GPU autoscaling on scheduled cost windows, fault tolerance (retries, dead-letter queues, fallback responses, exactly-once webhook delivery), and centralised logging and monitoring.
 - Applied Tip-Adapter (training-free CLIP adaptation) and DualCoOp (dual prompt tuning for multi-label recognition) on the SigLIP2 vision-language encoder with only 560 annotations, reaching >75% binary accuracy for heat-treatment classification and >70% average overall F1 on 28-type inclusion recognition.
 - Applied Segment Anything 2 with visual prompting for zero-shot gemstone masking and tracking in video, reaching J&F >85%.
 - Deployed MiniMax-Remover to remove handlers and inpaint occluded regions in video, reaching >8.1 visual quality and >95% success rate as judged by Gemini 2.5 Pro under an LLM-as-a-judge protocol.
 - Used Gemini 2.5 Pro with prompt chaining and reflection plus few-shot prompting for gemstone-versus-background detection (>85% average precision at IoU 0.75), and Gemini 2.5 Flash image generation for stain removal (DINOv2 similarity >90%).
 - Raised colour-grading performance to 90% F1 via semi-supervised learning with high-confidence pseudo-labelling on unlabelled data, and applied Cleanlab confident learning to reduce label noise.
 - Led adoption of Feast as a feature store and introduced MLflow Projects with a factory pattern, centralising feature management and standardising training execution for reproducibility and reuse.
- Ideagen ICT Sdn. Bhd.**, Senior Machine Learning Engineer Kuala Lumpur, Malaysia
Sept 2022 – Apr 2023
8 months
- Implemented and fine-tuned Microsoft's Table Transformer on in-house datasets, improving table parsing accuracy by at least 5%.
 - Optimised the PDF parsing pipeline in Ideagen AuditXprt using AWS Textract with document layout analysis (pdfminer, deep layout parser), improving parsing accuracy by 40% and speed by 250%.
 - Used MLflow for experiment tracking and as a centralised model registry, improving deployment efficiency by 30% across ML and software engineering.
 - Introduced Docker Compose for model inference APIs and Selenium regression testing for the PDF parsing tool, cutting debugging time by 30%.

Holmusk, Senior Data Scientist

- Directed a 20-person multi-disciplinary team to set the technical vision for an MVP, hitting each milestone two weeks ahead of schedule.
- Built hospitalisation and self-harm risk prediction models with a data enrichment method, reaching AUROC of at least 0.75.
- Engineered a calibration method for risk value adjustment across mental healthcare centres.
- Deployed MLflow as the team's MLOps platform and standardised post-training model evaluation reporting.

Singapore
Jan 2022 – Aug 2022
8 months

Holmusk, Data Scientist

- Launched customised Food Tracker AI into AIA Vitality across five countries, serving over 80k users and scoring over 2 million images.
- Developed lightweight transfer-learning models reaching >92% food detection and >90% food scoring accuracy.
- Implemented continual learning with elastic weight consolidation and rehearsal, enabling model refinement within 30 minutes post-launch.
- Pioneered an AWS SageMaker MLOps framework and doubled training throughput using multi-GPU EC2 instances.

Singapore
Mar 2019 – Dec 2021
2 years 10 months

Celcom Axiata Berhad, Data Scientist

- Built customer lifecycle segmentation for monetisation and defection probability.
- Developed a cross-selling model over sequential purchase behaviour, exceeding campaign revenue growth targets by 5%.
- Built an uplift model for promotion response likelihood, cutting message cost by 10%.

Kuala Lumpur, Malaysia
Sept 2017 – Feb 2019
1 year 6 months

Research Projects

Learning Self-Supervised Visual Representations for Fine-grained Maritime Vessel Recognition

MSc dissertation, Nanyang Technological University.

- Curated over 3.1M vessel images across 21 fine-grained categories via a data-sourcing pipeline, deduplicated with a Self-Supervised Copy Detection model.
- Built supervised Vision Transformer baselines (ViT-B/L/H) and compared self-supervised paradigms including MAE, DINOv2, and I-JEPA.
- Addressed class imbalance with balanced sampling and logit-adjusted loss.
- Reached 98.82% test accuracy over 17 in-distribution vessel classes and 100% over 4 out-of-distribution classes, showing self-supervised pre-training generalises better than supervised-only baselines.

Application of the Bees Algorithm to Improve the Performance of Flywheel Energy Storage

BEng final year project, Universiti Putra Malaysia.

- Optimised composite flywheel rotor geometry with the Bees Algorithm under the Tsai-Hill failure criterion to maximise energy storage while minimising mass.
- Validated top designs with finite element analysis in ANSYS, improving usable energy by 23.5% and reducing rotor mass by 41.5% against the NASA G3 baseline.

Awards

NTU Research Scholarship

Nanyang Technological University, Graduate College.

Apr 2026

Skills

Research: Self-supervised and representation learning, vision-language models, continual learning, reinforcement learning, domain adaptation, transfer learning, model distillation, post-training calibration

Computer Vision: Image classification, object detection, semantic segmentation, visual prompting, zero-shot masking and tracking

NLP and LLMs: Prompt engineering, in-context learning, retrieval-augmented generation, semantic matching, named entity recognition, topic modelling

Languages and Libraries: Python, SQL, PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, pandas, NumPy, SciPy

MLOps and Infrastructure: MLflow, Feast, TensorFlow Extended, FastAPI, Redis, Docker, AWS (SageMaker, EC2, S3), GCP (Vertex AI, Compute Engine), Optuna, Ax

Certificates

DeepLearning.AI (Coursera): Machine Learning; Deep Learning; Natural Language Processing; Machine Learning Engineering for Production (MLOps); AI for Medicine; Mathematics for Machine Learning and Data Science; Practical Data Science on AWS; Generative AI with LLMs

Amazon Web Services (Coursera): AWS Fundamentals Specialization